# Toward Submeter Satellite Surface Topography and Vegetation Mapping Using LiDAR/RGB Constrained Generative Diffusion

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***Abstract*—Sensing the Earth's surface topography and vegetation (STV) structure is of critical importance for a myriad of scientific applications. STV metrology relies on light detection and ranging (LiDAR), radio detection and ranging (RADAR), stereophotogrammetry, or a combination of these remote sensing techniques. STV metrology, however, suffers from low spatial and height resolution or sparse coverage if LiDARs and stereophotogrammetry are deployed at orbital heights. Many scientific applications, such as bare Earth, cryosphere, and hydrology, require meter or submeter STV observables in spatial resolution with submeter vertical resolution. This work aims to overcome the STV resolution gap by using a simple observation system composed of an orbital compressive sensing (CS) LiDAR aided by high-resolution monocular RGB photography. The system first produces a super-resolved digital surface model (DSM) by fusing satellite CS LiDAR photon returns with monocular photography using an image-to-image (I2I) translation generative Brownian bridge diffusion model (BBDM). Subsequently, the low photon count LiDAR measurements together with the high-resolution DSM are then used in a constrained denoising diffusion probabilistic model (DDPM) to reconstruct super-resolved, wall-to-wall, and feature-rich hyperheight STV data cubes. This approach effectively enhances the resolution for satellite LiDAR imagery while reducing generative model hallucinations, thereby improving the reliability and utility of the resulting data products for Earth studies. The achievable spatial resolution depends on the monocular RGB imagery resolution, the photon density of the training point cloud, and the noise level in the LiDAR sensor.**

***Index Terms*—Canopy height model (CHM), compressive light detection and ranging (LiDAR), digital surface model (DSM), generative diffusion models, surface topography and vegetation (STV).**

## NOMENCLATURE

| | |
|---|---|
| AGL | Above ground level. |
| BBDM | Brownian bridge diffusion model. |
| CASALS | Concurrent artificially-intelligent spectrometry and adaptive LiDAR system. |
| CHM | Canopy height model. |
| CHP | Canopy height profile. |
| CS | Compressive sensing. |
| C-DDPM | Conditioned denoising diffusion probabilistic model. |
| CSI-DDPM | Concatenated side-information denoising diffusion probabilistic model. |
| DDPM | Denoising diffusion probabilistic model. |
| DSM | Digital surface model. |
| DTM | Digital terrain model. |
| GCP | Ground control point. |
| GEDI | Global ecosystem dynamics investigation. |
| G-LiHT | Goddard's LiDAR, hyperspectral, and thermal system. |
| GT | Ground truth. |
| HHDC | Hyperheight data cube. |
| I2I | Image-to-image. |
| ICESat-2 | Ice, Cloud, and Land Elevation Satellite-2. |
| LiDAR | Light detection and ranging. |
| MAE | Mean absolute error. |
| mse | Mean squared error. |
| NEON | National Ecological Observatory Network. |
| NOS | Novel observing system. |
| NSF | National Science Foundation. |
| PSNR | Peak signal-to-noise ratio. |
| RADAR | Radio detection and ranging. |
| SCBI | Smithsonian Conservation Biology Institute. |
| SERC | Smithsonian Environmental Research Center. |
| SIM-ID | Side-information-masked inpainting diffusion. |
| SSIM | Structural similarity index measure. |
| STV | Surface topography and vegetation. |
| UKFS | University of Kansas Field Station. |

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## I. INTRODUCTION

CHARACTERIZING the Earth's surface, whether geological, vegetative, submerged, or constructed, is crucial for understanding various processes vital to health and sustainability. Natural geological phenomena such as earthquakes, volcanoes, and landslides continuously reshape the Earth's crust. Vegetation, with its diverse structures, not only shapes

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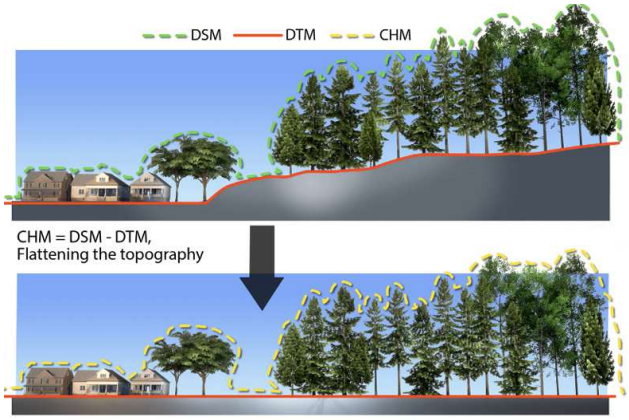



Fig. 1. STV elevation observables for an area containing a mix of vegetation and human structures. (Top) DSM and the DTM. (Bottom) Result of subtracting the DTM, obtaining the CHP. Reproduced from [7], Image credit: Chuck Carter, STV Study Team Report (2021).

habitats but also provides essential resources, deeply intertwined with Earth's evolving climate. The cryosphere, which includes ice and snow, is pivotal for freshwater supply. Measuring the Earth's STV structure is thus of paramount importance in addressing a myriad of scientific challenges [1].

Several STV observables are critical for characterizing Earth's STV structure. One critical observable is the DSM, which represents the elevation of the highest reflective surface, including elements above the bare Earth, such as vegetation and infrastructure. In forested areas, the DSM corresponds to the height of the top of the canopy (see Fig. 1). A second observable is the DTM, which captures the elevation of the bare Earth, excluding all features above the terrain (e.g., vegetation, trees, and buildings).

While these observables provide 2-D representations of elevation at specific points (e.g., top of the canopy and bare Earth), characterizing STV requires the 3-D distribution of aboveground elements and their variation across the landscape. Key vegetation attributes include canopy height, canopy cover, leaf area density profiles, and heterogeneity. To collect these types of features, STV metrology relies on remote sensing techniques such as LiDAR, RADAR, and stereophotogrammetry, where each method offers unique, complementary measurements critical for resolving Earth's structural variability.

Among these techniques, LiDAR and RADAR can penetrate vegetation foliage, capturing under-canopy structures, thus providing 3-D structural data. LiDAR sensor resolution and coverage depend on the sensor type and platform. For instance, airborne LiDARs such as NASA's Goddard LiDAR, Hyperspectral and Thermal Imager (G-LiHT), operate at low altitudes (~335 m), generating submeter resolution point clouds over relatively small areas. In contrast, satellite missions like NASA's GEDI [2] and ICESat-2 [3] operate at over 400-km altitude, providing lower resolution data but offering global coverage. G-LiHT generates dense, centimeter-resolution 3-D point clouds with high photon counts. In contrast, GEDI and ICESat-2 illuminate large ground footprints, producing height waveforms that characterize the vertical distribution and structure of vegetation of large aggregated areas. A sequence of these waveforms forms the CHP, which models the spatial distribution of vegetation elements along the satellite's track. GEDI, for instance, emits LiDAR pulses that create 25-m diameter footprints every 60 m along its track, with its swath consisting of eight CHPs spaced 600 m apart.

The limited coverage of airborne platforms and the sparsity of satellite measurements create significant gaps in STV data collection. To address this limitation, NASA's CASALS program proposes a novel approach centered on adaptive wavelength-scanning LiDARs. This system utilizes a wavelength-to-angle mapping dispersive grating to dynamically distribute laser beams across the observational track, enabling continuous footprint acquisition along the track and continuous CHPs across it. Particularly, CASALS's beam will be pointable to 1200 10-m footprints across a 7-km swath [4], achieving CHP measurements at approximately 6-m intervals across the swath.

A third technology to collect STV features is stereophotogrammetry which offers broad-area coverage and the ability to generate DSMs using passive optical imagery. However, unlike LiDAR, it cannot penetrate vegetation to capture subcanopy structures, limiting its effectiveness in densely forested regions. In addition, it may underestimate canopy height when tree crowns are narrower than the image pixel size [5]. The spatial resolution of stereophotogrammetry typically matches that of the original imagery, but its elevation accuracy varies significantly, depending on factors such as the base-to-height ratio and the presence of GCPs. High-resolution stereo imagery, such as that from WorldView-3, is often paired with multiple GCPs to improve elevation precision. While this enhances accuracy, it also increases operational complexity [6], reducing the efficiency of large-area mapping.

While LiDAR provides high-accuracy 3-D structural data, its coverage varies significantly depending on the sensor platform. Conversely, stereophotogrammetry enables broad-area DSM generation but does not capture subcanopy structures. These inherent tradeoffs between resolution and coverage underscore the challenges of relying on a single remote sensing technology to fully capture terrestrial structures.

One promising approach to advancing STV capabilities is the coordination of multiple sensors across various spatial and temporal scales, known as an NOS [8], [9], [10]. NOS leverages coordinated orbital and suborbital LiDAR, RADAR, and stereo-imaging sensors. Early efforts to fuse sensor data for NOS systems have demonstrated significant potential. For example, stereophotogrammetry has been used to generate DSMs that complement vegetation structure information from LiDAR [11]. Also, hyperspectral images have been combined with CASALS adaptive-wavelength LiDAR and deep learning methods to reconstruct information lost during photon propagation through the atmosphere [12], [13] or to optimize footprint distribution, reducing the costs of satellite LiDAR sampling [13].

The CASALS technology enables the continuous acquisition of measurements both along and across the track, effectively eliminating gaps in data collection. However,

despite this improvement, CASALS footprints still exhibit the coarse-resolution limitations inherent to satellite-based LiDAR systems. Previous works have aimed to enhance the resolution of satellite LiDAR measurements using deep learning techniques, leveraging super-resolution models to reconstruct finer spatial details from low-resolution inputs [14], [15]. While these methods have demonstrated potential in refining LiDAR-derived canopy structures and improving vegetation height estimations, the achievable super-resolution from purely coarse LiDAR measurements remains fundamentally limited. Moreover, deep learning models are susceptible to hallucinations, where the generated high-resolution details may not accurately represent the true underlying structure, leading to potential misinterpretations in scientific applications. Addressing these challenges requires robust model validation techniques and the integration of complementary multimodal high-resolution data sources to constrain and guide the super-resolution process effectively.

This article presents an NOS that aims to super-resolve coarse-resolution satellite LiDAR measurements, integrating high-resolution monocular optical imagery to overcome the limitations of current STV mapping technologies. To enhance the horizontal and vertical resolution of the measurements, this methodology leverages state-of-the-art generative artificial intelligence methodologies. Our method focuses on the reduction of the model's hallucinations to increase the usability of the generated imagery. The proposed approach enables the retrieval of STV observables with 1-m horizontal resolution and submeter vertical precision, which is critical for applications such as bare Earth and hydrology mapping [16]. The methodology consists of two sequential stages. In the first stage, high-resolution orbital RGB imagery—such as that collected by WorldView satellites [17] or Planet SkySat-C [18]—or suborbital data—such as that collected by NEON airborne observation platform (AOP) [19]—when available, is fused with coarse-resolution satellite LiDAR measurements to estimate a high-resolution DSM. In the second stage, the high-resolution DSM is used to derive a binary tensor mask, which serves as a structural constraint to improve the accuracy of the super-resolution process, reducing artifacts and hallucinations in the reconstructed elevation data. This integration of LiDAR and optical data enhances the spatial fidelity of STV measurements, overcoming the resolution limitations of standalone satellite LiDAR systems.

In this article, we build on foundational concepts in diffusion models to propose methodologies for super-resolving coarse-resolution DSMs by translating from RGB images. Furthermore, we define a new way to condition the HHDC reconstruction process by constraining the generative process on a proposed third-order tensor mask. The main contributions of this work can be summarized as follows.

1) Definition of a novel RGB-to-DSM translation task. We provide the formal methodology for translating a monocular RGB image into the corresponding high-resolution DSM using generative AI. Specifically, we apply BBDMs and evaluate its performance against other diffusion methods like DDPMs.
2) We propose a novel binary tensor mask derived from high-resolution DSMs. Unlike conventional DSMs, which represent elevation in a 2-D format, this mask is generated to describe the 3-D structure of the canopy within a 3-D tensor. This representation enhances the subsequent generative process of diffusion models by providing a more detailed and spatially accurate depiction of the terrain, improving the reliability of reconstructed HHDCs, and minimizing hallucinations.
3) Definition of a new conditioning mechanism for HHDC reconstruction that constrains the generative process by incorporating the proposed third-order tensor mask, effectively guiding the diffusion model to produce more accurate and realistic HHDC reconstructions.
4) Evaluation of side information integration in DDPMs: we conduct an extensive comparison of methods for incorporating side information into DDPMs for 3-D spaces modeled as tensors. Our findings demonstrate that the proposed methodology, which leverages advanced integration techniques, significantly outperforms multimodal approaches such as concatenation, yielding superior reconstructions from the same input data.

The remainder of this article is organized as follows. Section II provides background on satellite and compressed sensing LiDAR. Section III details the proposed methodology, while Section IV describes the dataset, training procedures, and hardware setup for reproducibility. Section V presents the evaluation metrics, experimental results, and performance analysis. Finally, Section VI summarizes the findings and outlines potential directions for future research.

## II. Satellite LiDAR Preliminaries

### A. LiDAR Waveform Sensing and The Effect of Orbit Altitude

The light propagation properties in the atmosphere and isotropic reflection on Earth's surfaces define the sensing mechanisms in compressive satellite LiDARs (CS-LiDARs). The forward imaging system $\mathcal{A}_{\mathbf{\Psi}}(\cdot)$ models photon pulse behavior, influencing the resolution limit of the system. It is characterized by the following aspects.

*1) Coded Illumination Pattern*, $\mathbf{\Psi}$, which defines the laser's spatial sampling and achievable resolution. Digital micromirror device or fast-tunable lasers with a dispersive grating can generate beam scanning over kilometers without mechanical parts, producing various laser footprints and spacing patterns [20], [21], [22].

*2) Photon Attenuation*, $\tau$, governs photon extinction as pulses travel from the Earth's surface to the sensor, caused by molecular absorption and particle scattering.

*3) Isotropic Surface Reflection*, $\rho$, determines energy loss due to photon absorption and dispersion by the Earth's surface. Reflection density decays with a factor of $z^2$, where $z$ is the distance from the target to the sensor [23].

The LiDAR equation

$$E_{\text{shot}} = E_{\text{det}} \cdot \frac{2\pi z^2}{QA\rho\tau^2} \tag{1}$$

relates the transmitted pulse energy $E_{\text{shot}}$ to the detected photon distribution energy $E_{\text{det}}$. Here, $Q$ is the detector's

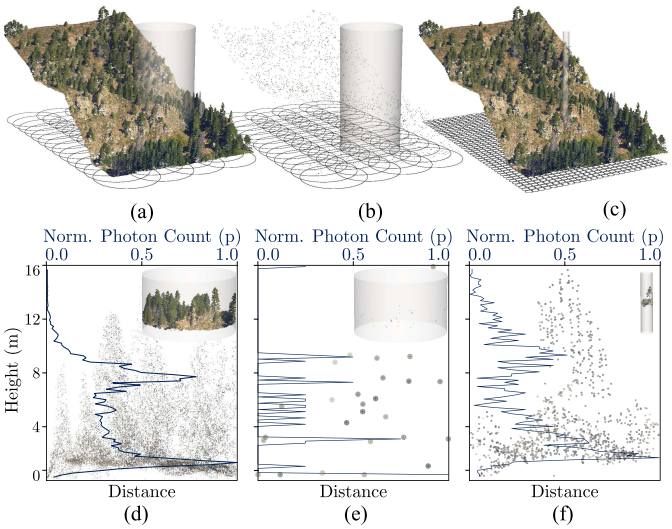



Fig. 2. (a) NEON high-resolution point cloud captured at 1-km AGL. (b) Photon-starved point cloud captured at 500 km in CS-LiDAR. (c) NEON high-resolution point cloud captured at 1-km AGL. (d) and (e) Normalized height-histogram waveforms for the coarse-resolution footprint captured at 1 and 500 km, respectively. (f) Normalized height-histogram waveform for the fine-spatial-resolution footprint captured at 1 km.

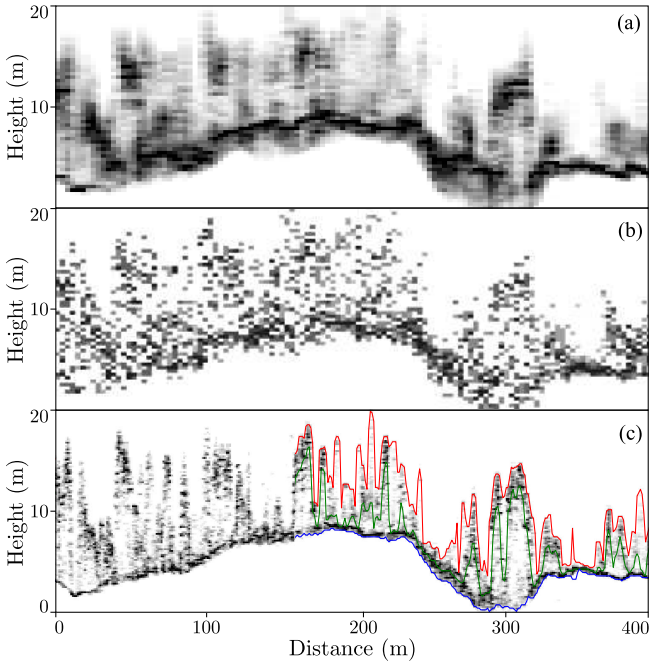



Fig. 3. (a) and (b) CHPs formed by concatenating a sequence of coarse-resolution footprints along-track path captured at 1 and 500 km, respectively. (c) CHP is formed by concatenating a sequence of high-resolution footprints along track captured at 1-km AGL. Red, green, and blue lines show the 98th, 50th, and 2nd height waveform percentiles, respectively.

quantum efficiency, and $z$ is the LiDAR altitude, which significantly impacts the detected energy of the returning pulse [24]. The telescope area $A$ determines the diffraction limit and resolution. However, spaceborne LiDAR resolution is power-limited, and the effects of $\tau$ and $\rho$ on returned photons are amplified by the satellite's distance from Earth. Although thousands of photons are emitted per pulse, just a couple of dozen are detected back.

Fig. 2 illustrates this phenomenon using data from a portion of the Rocky Mountain National Park in Colorado. Fig. 2(a) shows the high-resolution, high-density point cloud of the scene, acquired with NEON's AOP LiDAR [19] at 1000-m AGL. The figure also highlights a cylindrical volume (footprint) simulating the volume illuminated by a large LiDAR laser pulse. Within this footprint, thousands of individual photons are recorded, and their time-of-flight measurements generate the height profile shown in blue in Fig. 2(d), depicting the count of photon reflections from various heights within the footprint. This profile is commonly referred to as the height *waveform* and can be specified by a histogram representing the sum of all photons within specific height ranges. We can simulate the same region being measured from a satellite orbiting at 500-km AGL, in that case, most of the photons are extinguished as they travel through the atmosphere, with only a small fraction recorded, as shown in Fig. 2(b). The few points inside the footprint volume in Fig. 2(b) generate the sparse height histogram *waveform* shown in Fig. 2(e) in blue. Previous work [13], [25] focused on simulating satellite measurements [see Fig. 2(b)] and reconstructing them using deep learning methods to match the photon density of airborne data [see Fig. 2(a)] while preserving spatial resolution and footprint size. This work extends these methods by reconstructing the features of full photon-count waveforms and also reducing the footprint diameter to as small as 1.41 m. As shown in Fig. 2(c), reducing footprint size helps preserve more terrain details by minimizing the aggregated area, which is reflected in the corresponding waveform as shown in Fig. 2(f).

A sequence of height waveforms can be concatenated to form a CHP, a widely used representation of waveform LiDAR data [10], [26]. CHPs provide valuable insights into landscape features such as topography and forest density, as well as vegetation characteristics like height distribution, density, and health. Fig. 3 shows three simulated CHPs with different characteristics: Fig. 3(a) presents an example CHP simulating data acquisition at 1-km AGL over a section of Rocky Mountain National Park, where each column corresponds to a 10-m diameter footprint, with darker regions indicating denser vegetation (higher photon concentration). However, satellite LiDAR systems generate photon-starved CHPs that lack finer terrain details due to lower photon return rates, like the simulation in Fig. 3(b). Increasing the resolution—by reducing footprint diameters and spacing between them—preserves more structural details, as demonstrated in Fig. 3(c), where a CHP is constructed using 1-m diameter footprints spaced 1 m apart.

Key percentiles of height waveforms provide critical insights into terrain and vegetation structure. As illustrated in Fig. 3(c), the 2nd percentile (blue line) represents the bare Earth elevation, forming the DTM, while the 98th percentile (red line) marks the top of the canopy, defining the DSM. The CHM, computed as DSM—DTM, quantifies tree height relative to the ground [25]. The DSM captures the elevation of the highest reflective surface, including vegetation and infrastructure, whereas the DTM isolates the underlying terrain, filtering out features above the ground (e.g., trees and

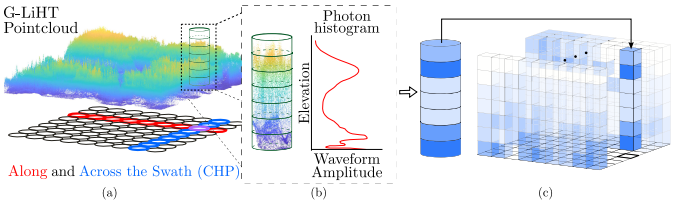



Fig. 4. HHDC: (a) tube-shaped illuminated footprint, (b) histogram of heights within the footprint, and (c) concatenated height histograms building the 3-D HHDC.

buildings). The CHM, derived from their difference, provides a direct measurement of canopy height, offering valuable information for forest structure analysis and ecological studies (see Fig. 1). Fig. 3(c) also shows the 50th percentile (green line), which is used to estimate vegetation parameters like the above-ground biomass.

### B. Hyperheight Data Cubes

CHPs are 1-D arrays of concatenated height waveforms. These arrays can be extended to two dimensions by concatenating continuous CHPs in the direction across the track of the LiDAR sensor's movement. This concatenation results in a 3-D representation known as the HHDC [25]. Similar to the CHPs, each vertical element in the HHDC represents the height distribution within a specific footprint volume. CHPs can be extracted as vertical slices from the HHDC, either along or across the track, and height percentiles from the HHDC can generate 2-D maps for the DSMs, DTMs, and CHMs, enhancing data interpretation [25].

The HHDC is modeled as a 3rd-order tensor $\mathcal{X} \in \mathbb{R}^{m\times n\times c}$, where $m$ and $n$ are the number of footprints across and along the track, respectively, and $c$ is the number of height bins. The *horizontal resolution* of an HHDC is defined as $\Delta m \times \Delta n$ where $\Delta m$ and $\Delta n$ are the distances between adjacent footprints' centers across and along the track, respectively. The *vertical resolution* refers to the granularity of height intervals, defined by the bin size $\Delta c$.

*1) HHDC Simulation:* Airborne and satellite HHDCs can be simulated from high-photon-density point clouds. Fig. 4 shows the simulation process. The simulation begins by overlaying an $m \times n$ grid onto the point cloud's area [see Fig. 4(a)], where $m$ and $n$ are the number of tube-shaped footprints in the direction across and along the track, respectively. Fig. 4(a) highlights one of the footprints, representing the area illuminated by a single LiDAR laser shot. The points within this footprint are used to compute a histogram of point heights. For high-resolution airborne HHDC simulations, the histogram includes all points within the footprint [see Fig. 4(b)], while in coarse-resolution satellite simulations, a randomly selected subset is used, following the Gaussian illumination model described in [25]. The resulting histogram serves as the photon-counting signal for the footprint, assigned to its position on the grid [see Fig. 4(c)]. Each voxel in the HHDC represents the photon count within a cylindrical region defined by the bin height and the footprint diameter. In Fig. 4(c), the photon count is visually represented by shades of blue, with darker tones indicating higher counts.

## III. Super Resolution HHDC via Constrained Generative Diffusion Models

Various approaches have been developed to enhance both the vertical and horizontal resolution of satellite LiDAR measurements [13], [25]. Early approaches focused on increasing the horizontal resolution of HHDCs using deep convolutional neural networks with adversarial loss functions [15]. More recent techniques leverage denoising diffusion models to achieve more realistic reconstructions while simultaneously improving both vertical and horizontal resolution [14], [27]. However, as the resolution of reconstructions increases, hallucinations and artifacts often emerge, limiting the practical applicability of these methods.

This article presents a new methodology that takes advantage of the high-resolution information provided by RGB cameras and the capabilities of generative models to enhance the resolution of HHDCs while maintaining high fidelity and minimizing hallucinations. The proposed methodology consists of two stages: in stage one, a high-resolution RGB image is used to estimate a high-resolution DSM. In stage two, the high-resolution DSM is used to generate a third-order binary tensor mask, which guides and constrains a conditioned DDPM [28] to generate the high-resolution HHDC. The proposed methodology provides a more appropriate approach for incorporating side information into the HHDC super-resolution process, enabling better control and guidance to the DDPM generation process compared to solely concatenating the side information, as shown in Section V.

For a given region, let $\boldsymbol{x}_{\text{RGB}} \in \mathbb{R}^{m\times n\times 3}$ represent the high-resolution RGB image. The corresponding $\boldsymbol{x}_{\text{DSM}} \in \mathbb{R}^{m\times n\times 1}$ represents the region's DSM at the same horizontal resolution as $\boldsymbol{x}_{\text{RGB}}$. Equivalently, $\check{\boldsymbol{x}}_{\text{DSM}} \in \mathbb{R}^{\alpha m\times \beta n\times 1}$ represents the region's coarse-resolution DSM where $\alpha, \beta < 1$ are the downsampling factors for each dimension.

In stage one, the high-resolution RGB image $\boldsymbol{x}_{\text{RGB}}$ is used to estimate the corresponding high-resolution DSM $\boldsymbol{x}_{\text{DSM}}$ through an I2I translation diffusion model. Estimating the DSM from an RGB image alone is challenging due to the nonbijective nature of the mapping [29]; for instance, small changes in the elevation map may not substantially affect the RGB image. To overcome this challenge and enable accurate reconstruction of absolute altitudes, the coarse-resolution DSM $\check{\boldsymbol{x}}_{\text{DSM}}$ is incorporated into the input of the I2I translation process. This is achieved through channel concatenation

$$\mathring{\boldsymbol{x}}_{\text{RGB}} = \left[\left(\mathcal{I}_b\left(\check{\boldsymbol{x}}_{\text{DSM}}\right)\right), \boldsymbol{x}_{\text{RGB}}\right] \tag{2}$$

where $\mathring{\boldsymbol{x}}_{\text{RGB}} \in \mathbb{R}^{m\times n\times 4}$ is the resulting *altitude-aware* RGB image, and $\mathcal{I}_b(\cdot) : \mathbb{R}^{\alpha m\times \beta n\times 1} \to \mathbb{R}^{m\times n\times 1}$ denotes the bilinear interpolation function used to resize $\check{\boldsymbol{x}}_{\text{DSM}}$ to match the size of $\boldsymbol{x}_{\text{RGB}}$. The altitude-aware RGB image $\mathring{\boldsymbol{x}}_{\text{RGB}}$ is then used as input to estimate the high-resolution DSM $\boldsymbol{x}_{\text{RGB}}$ throughout the I2I translation diffusion model.

Recent advancements in diffusion models have significantly improved I2I translation by leveraging a probabilistic framework for high-quality image generation [30], [31]. These methods build upon the standard DDPMs [32], which involve a forward diffusion process that progressively adds Gaussian noise to the input image, followed by a reverse process that

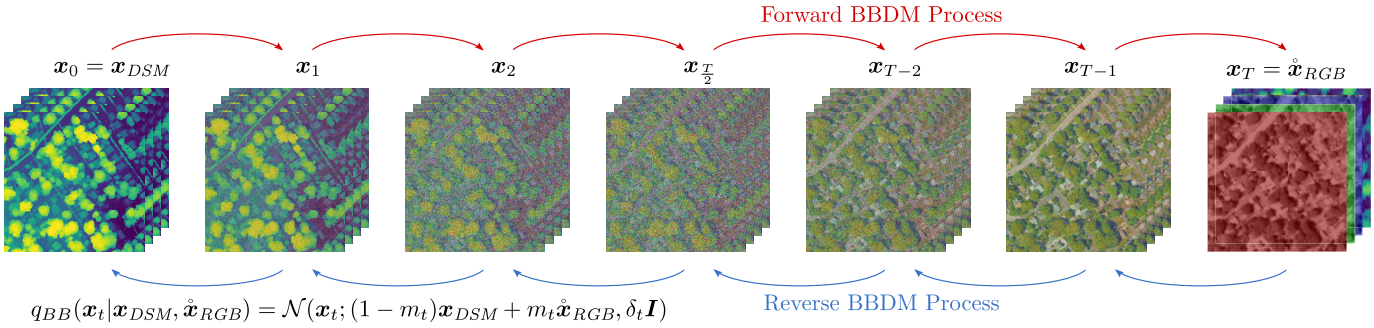



Fig. 5. Forward and reverse processes in the BBDM. The $T$-step forward process begins with the high-resolution DSM image and progresses to generate high-resolution RGB and coarse-resolution DSM images concatenated along the channel dimension. The noise level increases from step $t = 0$ to $t = T/2$, then decreases until $t = T$. In the reverse process, starting at $t = T$, a neural network iteratively reconstructs the high-resolution DSM image by sequentially reversing each diffusion step.

reconstructs the target image while conditioning on the source image.

Diffusion models have been successfully applied to various I2I translation tasks, such as medical image synthesis [33], satellite-to-map conversion [34], and artistic style transfer [35]. The recently proposed BBDM [36] refines this approach by incorporating the clean source image directly in the forward process, making it particularly well-suited for I2I translation tasks. This advancement reduces noise distortion in the conditioning process and enhances the quality of fine-grained details in the generated images.

*1) BBDM Forward Process:* The BBDM forward process [36] progressively diffuses the high-resolution DSM, $\boldsymbol{x}_{\text{DSM}}$, into the altitude-aware colorized image, $\mathring{\boldsymbol{x}}_{\text{RGB}}$, over $T$ steps. This process follows a Gaussian distribution $q_{\text{BB}}(\boldsymbol{x}_t|\boldsymbol{x}_{\text{DSM}}, \mathring{\boldsymbol{x}}_{\text{RGB}})$ defined as

$$q_{\text{BB}} = \mathcal{N}\left(\boldsymbol{x}_t; (1-m_t)\,\boldsymbol{x}_{\text{DSM}} + m_t\mathring{\boldsymbol{x}}_{\text{RGB}}, \delta_t \boldsymbol{I}\right) \tag{3}$$

where $t$ represents the current step, $m_t = t/T$ is a weighting factor that interpolates between $\boldsymbol{x}_{\text{DSM}}$ and $\mathring{\boldsymbol{x}}_{\text{RGB}}$, and $\delta_t$ is the step variance, defined as $\delta_t = 2s(m_t - m_t^2)$. The scalar $s$ is a hyperparameter that adjusts the magnitude of variance to control sampling diversity.

The forward process begins at $t = 0$, where the latent variable $\boldsymbol{x}_0$ equals the high-resolution DSM, $\boldsymbol{x}_{\text{DSM}}$, with variance $\delta_0 = 0$. At $t = T$, the latent variable $\boldsymbol{x}_T$ equals the altitude-aware colorized image, $\mathring{\boldsymbol{x}}_{\text{RGB}}$, with variance $\delta_T = 0$. The variance, $\delta_t$, evolves nonlinearly: it increases during the first half of the process, peaking at $\delta_{T/2}$ with a value of $1/2s$, and then decreases to zero in the second half.

The forward process is illustrated in Fig. 5 for a 384 $\times$ 384 m region. In the beginning of the process ($t = 0$), the latent variable $\boldsymbol{x}_0$ is equal to the DSM $\boldsymbol{x}_{\text{DSM}}$. Over the following steps, $\boldsymbol{x}_t$ slowly incorporates components of $\mathring{\boldsymbol{x}}_{\text{RGB}}$ and noise. During the first half of the process, the noise variance $\delta_t$ increases, peaking at $\delta_{T/2}$ where the influence of $\boldsymbol{x}_{\text{DSM}}$ and $\mathring{\boldsymbol{x}}_{\text{RGB}}$ is the same. In the second half, both the variance and the influence of $\boldsymbol{x}_{\text{DSM}}$ diminish. By the end of the process ($t = T$), the noise variance is zero, and the latent variable $\boldsymbol{x}_T$ equals the altitude-aware colorized image, $\mathring{\boldsymbol{x}}_{\text{RGB}}$.

### A. Stage One: High-Resolution DSM Estimation

*1) BBDM Reverse Process:* The BBDM reverse process begins with the altitude-aware colorized image, $\mathring{\boldsymbol{x}}_{\text{RGB}}$, and reconstructs the high-resolution DSM, $\boldsymbol{x}_{\text{DSM}}$, over $T$ steps. At each step $t$, the model predicts the latent variable $\boldsymbol{x}_{t-1}$ for the previous timestep using a parameterized Gaussian distribution [36]

$$p_\theta\left(\boldsymbol{x}_{t-1}|\boldsymbol{x}_t, \mathring{\boldsymbol{x}}_{\text{RGB}}\right) = \mathcal{N}\left(\boldsymbol{x}_{t-1}; \boldsymbol{\mu}_\theta\left(\boldsymbol{x}_t, t\right), \tilde{\delta}_t \boldsymbol{I}\right) \tag{4}$$

where $\boldsymbol{\mu}_\theta(\boldsymbol{x}_t, t)$ is the predicted mean produced by a neural network $\boldsymbol{\mu}_\theta$ with parameters $\theta$, and $\tilde{\delta}_t$ is the variance for timestep $t-1$. As recommended in [28] and [36], in this implementation we use a U-Net-based neural network for $\boldsymbol{\mu}_\theta$ (see Fig. 6). The variance in the reverse process is defined as [36]

$$\tilde{\delta}_t = \frac{\delta_{t|t-1} \cdot \delta_{t-1}}{\delta_t} \tag{5}$$

$$\delta_{t|t-1} = \delta_t - \delta_{t-1}\frac{(1-m_t)^2}{(1-m_{t-1})^2} \tag{6}$$

where $\tilde{\delta}_t$ represents the total variance at timestep $t-1$, and $\delta_{t|t-1}$ is the incremental variance between steps $t$ and $t-1$. The neural network learns to denoise and progressively reconstruct the DSM, $\boldsymbol{x}_{\text{DSM}}$, by leveraging both the noisy intermediate variable, $\boldsymbol{x}_t$, and the altitude-aware colorized image, $\mathring{\boldsymbol{x}}_{\text{RGB}}$.

### B. Stage Two: Constrained Diffusion Model

For a given region, let $\mathcal{X} \in \mathbb{R}^{m\times n\times c}$ represent the high-resolution HHDC captured by a LiDAR system at low altitudes. The corresponding $\check{\mathcal{X}} \in \mathbb{R}^{\alpha c\times \beta m\times \gamma n}$ denotes the coarse-resolution, photon-starved HHDC for the same region captured by a satellite system, where $\alpha, \beta, \gamma < 1$ are the downsampling factors for each dimension. The RGB image $\boldsymbol{x}_{\text{RGB}} \in \mathbb{R}^{m\times n\times 3}$ contains colorized information about the canopy in $\mathcal{X}$, and the DSM $\boldsymbol{x}_{\text{DSM}} \in \mathbb{R}^{m\times n\times 1}$ is derived from $\mathcal{X}$.

For stage two, we propose the SIM-ID method, which synthesizes the super-resolved HHDC $\hat{\mathcal{X}} \in \mathbb{R}^{m\times n\times c}$ by leveraging the high-resolution DSM $\boldsymbol{x}_{\text{DSM}}$ and the coarse-resolution HHDC $\check{\mathcal{X}}$.

The first step in SIM-ID involves embedding the high-resolution canopy structure information from $\boldsymbol{x}_{\text{DSM}}$ into a

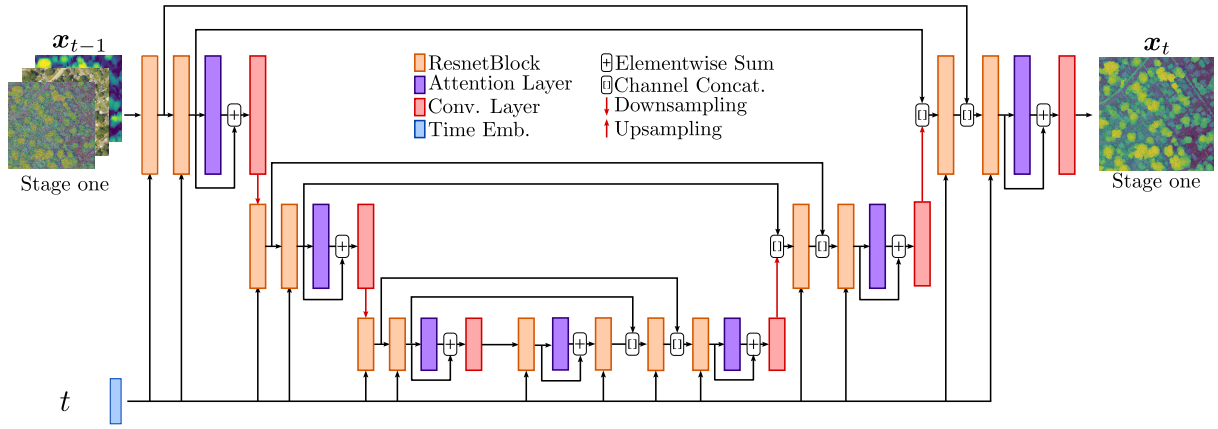



Fig. 6. U-Net-based neural network architecture for reverse diffusion processes. This neural network is utilized in both DDPM and the BBDM to denoise intermediate diffusion steps during the reverse process. The network takes as input the noisy image at timestep $t$, concatenated along the channel dimension (Channel Concat.) with the conditioning information and the scalar timestep value $t$ as a time embedding (Time Emb.). This architecture consists of an encoder–decoder structure with skip connections to preserve spatial details. The encoder progressively extracts hierarchical features through convolutional layers, residual convolutional blocks, and vision attention layers, while the decoder reconstructs the denoised image by merging these features with the same type of layers. This design enables effective modeling of multiscale features, ensuring precise denoising across all timesteps [28].

third-order binary tensor mask, $\mathcal{M} \in \{0,1\}^{m\times n\times c}$. The mask is defined as

$$\mathcal{M}^{i,j,k} = \begin{cases} 1, & k \geq \boldsymbol{x}_{\text{DSM}}^{i,j} \\ 0, & k < \boldsymbol{x}_{\text{DSM}}^{i,j} \end{cases} \quad i = 0,\ldots,m,\ j = 0,\ldots,n \tag{7}$$

where each column $\mathcal{M}^{i,j}$ contains 0's below the canopy and 1's above it. This mask, which shares the same dimensions as the high-resolution HHDC $\mathcal{X}$, outlines the 3-D canopy structure at high resolutions. The mask serves two primary purposes: 1) it provides the model with direct access to the canopy structure within the same domain as the HHDC, avoiding the need to learn to translate 2-D DSM or RGB data into a 3-D representation; and 2) it prevents hallucinations, particularly in the canopy structure, by constraining the model's output.

An example slice of a mask $\mathcal{M}$ is shown in Fig. 7(a), corresponding to the CHP in Fig. 7(b). The mask was generated from the DSM in Fig. 7(c), where the red line indicates the slice location. In Fig. 7(a), the black values (representing 1's) outline the canopy's structure.

The second step in SIM-ID combines the mask $\mathcal{M}$ with the coarse-resolution HHDC $\check{\mathcal{X}}$. To ensure compatibility, $\check{\mathcal{X}} \in \mathbb{R}^{\alpha m\times \beta n\times \gamma c}$ is resized to $\mathbb{R}^{m\times n\times c}$ using trilinear interpolation. The resized HHDC is then elementwise-multiplied by the complement of the mask $(1-\mathcal{M})$ to produce the combined tensor

$$\check{\mathcal{X}}_{\mathcal{M}} = \mathcal{I}_t\left(\check{\mathcal{X}}\right) \odot (1-\mathcal{M}) \tag{8}$$

where $\check{\mathcal{X}}_{\mathcal{M}} \in \mathbb{R}^{m\times n\times c}$ is the combined tensor, $\mathcal{I}_t(\cdot) : \mathbb{R}^{\alpha m\times \beta n\times \gamma c} \to \mathbb{R}^{m\times n\times c}$ denotes trilinear interpolation, and $\odot$ represents elementwise multiplication. Fig. 7(d) illustrates a CHP from the resized coarse-resolution HHDC, which is inherently blurry and noisy. By multiplying with $(1-\mathcal{M})$, the canopy noise is reduced, and fine-scale canopy structure is integrated, as shown in Fig. 7(e).

The third step in SIM-ID synthesizes the super-resolved HHDC $\hat{\mathcal{X}}$ from $\check{\mathcal{X}}_{\mathcal{M}}$ using an inpainting DDPM. The forward process in SIM-ID, as in the original DDPM [28], progressively adds Gaussian noise to the input $\mathcal{X}$ over $T$ steps, following:

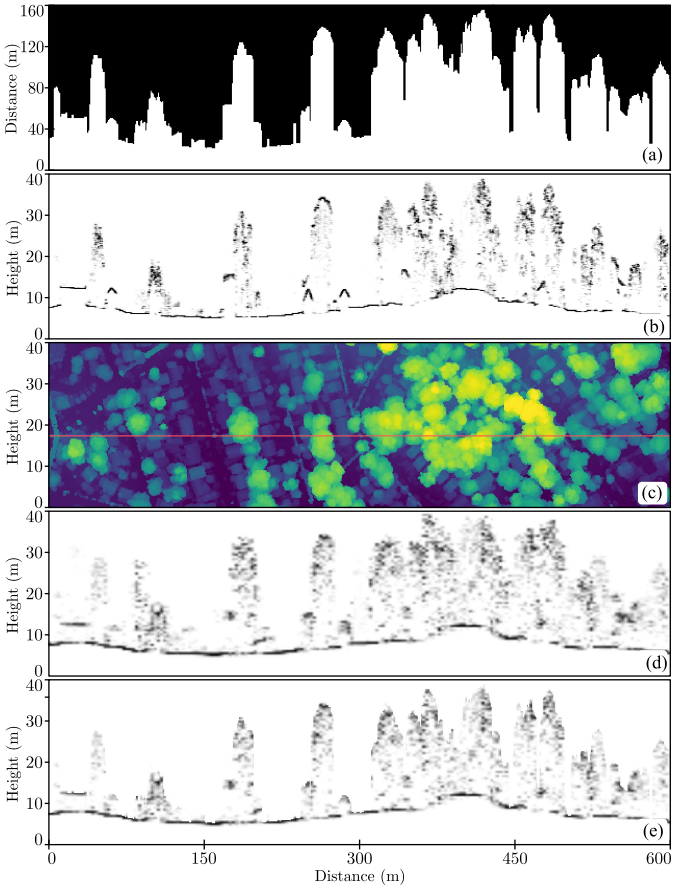



Fig. 7. (a) Slice from a third-order binary tensor mask, $\mathcal{M}$. (b) High-resolution CHP for the same region as in (a). (c) DSM at 1-m horizontal resolution. The red line shows the vertical slice selected for all the profiles. (d) Photon-starved coarse-resolution CHP interpolated to match the size of (b). (e) Element-wise product of (a) and (d) to induce high-resolution canopy structure information into the coarse-resolution LiDAR measurement.

$$q\left(\mathcal{X}_t|\mathcal{X}_{t-1}\right) = \mathcal{N}\left(\mathcal{X}_t;\ \sqrt{\alpha_t}\mathcal{X}_{t-1}, (1-\alpha_t)\,\mathbf{I}\right) \tag{9}$$

where $\mathcal{X}_t \in \mathbb{R}^{m\times n\times c}$ is the noisy HHDC at time step $t$, $\alpha_t \in (0,1)$ determines the noise addition rate, and $\mathbf{I}$ is the identity covariance matrix. By the end of the forward process, the cube becomes fully noisy, following a standard Gaussian distribution.

The reverse process iteratively denoises $\mathcal{X}_T$ over $T$ steps to recover $\mathcal{X}$, conditioned on $\check{\mathcal{X}}_{\mathcal{M}}$ following:

$$p\left(\mathcal{X}_{t-1}|\mathcal{X}_t, \check{\mathcal{X}}_{\mathcal{M}}\right) = \mathcal{N}\left(\mathcal{X}_{t-1}; \boldsymbol{\mu}_\theta\left(\mathcal{X}_t, t, \check{\mathcal{X}}_{\mathcal{M}}\right), \sigma_t^2\mathbf{I}\right) \tag{10}$$

where $p(\hat{\mathcal{X}}_{t-1}|\hat{\mathcal{X}}_t, \check{\mathcal{X}}_{\mathcal{M}})$ is the conditional probability of the denoised tensor at step $t-1$, $\boldsymbol{\mu}_\theta(\hat{\mathcal{X}}_t, t, \check{\mathcal{X}}_{\mathcal{M}})$ is the predicted mean, and $\sigma_t^2\mathbf{I}$ is the fixed diagonal covariance matrix. The predicted mean, $\boldsymbol{\mu}_\theta(\hat{\mathcal{X}}_t, t, \check{\mathcal{X}}_{\mathcal{M}})$, is parameterized by a U-Net-based neural network $\boldsymbol{\mu}_\theta$, which depends on the current noisy tensor $\hat{\mathcal{X}}_t$, the time step $t$, and the conditioning tensor $\check{\mathcal{X}}_{\mathcal{M}}$. The conditioning tensor is concatenated to $\hat{\mathcal{X}}_t$ along the channel dimension and passed as input to the network. Fig. 6 illustrates this U-Net architecture.

Voxels with value 1 in the binary mask $\mathcal{M}$ represent regions above the canopy and are expected to have zero values in the super-resolved HHDC $\hat{\mathcal{X}}$. This setup aligns with an inpainting task, where the model generates information only for the unknown regions of the cube (voxels where $\mathcal{M} = 0$). To enforce consistency with the canopy structure, the generated cube is modified at each step $t$ of the reverse process to force the inclusion of the known canopy structure. This modification helps guide the model to produce reconstructions that respect the high-resolution canopy structure and that harmonize with the known information.

Following the inpainting strategy in [37], two cubes are defined at each step $t$: $\hat{\mathcal{X}}_{t-1}^{\text{known}} \in \mathbb{R}^{m\times n\times c}$, containing the known canopy information (voxels above the canopy set to zero), $\hat{\mathcal{X}}_{t-1}^{\text{unknown}} \in \mathbb{R}^{m\times n\times c}$, output of the model for the noisy tensor $\hat{\mathcal{X}}_t$. At each reverse step, the cube is updated as

$$\hat{\mathcal{X}}_{t-1} = \hat{\mathcal{X}}_{t-1}^{\text{known}} \odot \mathcal{M} + \hat{\mathcal{X}}_{t-1}^{\text{unknown}} \odot (1 - \mathcal{M}) \tag{11}$$

where $\odot$ denotes elementwise multiplication. This modification ensures the inclusion of the known canopy structure in $\hat{\mathcal{X}}_{t-1}$.

By iteratively following this process across $T$ steps, the cube is fully denoised, yielding the super-resolved HHDC $\hat{\mathcal{X}}$.

## IV. Dataset Description and Training Setup

### A. HHDC Dataset

To train, validate, and test the proposed methodology, an HHDC dataset was generated by simulating satellite and airborne HHDC samples from airborne-collected point clouds. The satellite HHDCs simulate coarse-resolution data, while the airborne HHDCs represent high-resolution measurements. The satellite HHDC footprints were designed to replicate the resolution and properties of CASALS's footprints, with a horizontal spacing of 3 m along-track and 6 m across-track, a diameter of 10 m, and a vertical resolution of 0.5 m. To simulate photon loss due to sensor distance, each footprint contained approximately 20 photons, randomly sampled following a Gaussian distribution [25]. In contrast, the airborne HHDC footprints were configured with finer characteristics, featuring a horizontal spacing of 1 m in both along-track and across-track directions, a vertical resolution of 0.25 m, and a diameter of 1.41 m, ensuring no gaps between footprints. The airborne HHDC footprints retained the full photon content from the high-density point cloud.

The HHDC dataset was simulated following the methodology described in HHDC simulation and using real LiDAR point clouds collected by NEON's AOP LiDAR [19]. NEON's AOP operates at an altitude of 1000-m AGL and a velocity of 100 knots, capturing up to 40 points per square meter. NEON's AOP conducts surveys across various locations in the United States, covering a minimum area of 10 × 10 km per site. Data from three NEON sites were used: the UKFS (39.04°N, 95.19°W); the SCBI in Virginia (38.88°N, 78.15°W); and the SERC in Maryland (38.89°N, 76.58°W). All data were collected in May 2022.

The dataset was divided into approximately 20 000 subregions, each covering an area of 96 × 96 m, with up to 10% overlap between adjacent subregions. For each subregion, both coarse-resolution and high-resolution HHDCs were simulated. From these, 85% of the subregions were randomly allocated for training, 10% for validation, and 5% for testing. A separate 1 × 1 km subregion from SERC, containing notable features, was excluded from the main dataset and reserved for evaluation. Some regions contain elements, like water concentrations, that do not return photons at the laser's wavelength, creating holes in the point cloud and empty footprints in the HHDCs. These empty footprints introduce discontinuities and distort the true data distribution. Therefore, HHDCs with a percentage of empty footprints higher than 10% were filtered out from the dataset. After this filtering, approximately 18 000 pairs were kept.

The proposed methods require HHDC values to be within the range $[-1, 1]$. To achieve this, each footprint waveform is normalized using min–max scaling, followed by a transformation to the desired range. Even though higher photon counts in the HHDC footprints enable the collection of high-resolution features, these features are effectively preserved within the normalized range. While the model cannot reconstruct the absolute photon count—due to factors unrelated to the data distribution—it successfully learns to generate those high-resolution features in the normalized $[-1, 1]$ range.

### B. RGB Imagery Dataset

An RGB image dataset was created to provide side information for the airborne HHDCs, ensuring that each simulated airborne HHDC is paired with a corresponding high-resolution RGB image. In addition to the LiDAR sensor, NEON's AOP is equipped with a digital camera [38] that captures RGB imagery simultaneously over the same regions as the LiDAR. The RGB images have a horizontal resolution of up to 0.1 × 0.1 m and are mapped to the same geographic projection as the LiDAR data via orthorectification. To align with the airborne HHDCs, the RGB images were interpolated to match their horizontal resolution.

### C. DSM Dataset

The DSM dataset was derived from the HHDC dataset to train, validate, and test stage one of the proposed methodology. For each 96 × 96 m subregion in the HHDC dataset, airborne and satellite versions of the DSM were generated, creating 18 000 pairs of coarse and high-resolution DSMs.

### D. Training Setup

The training for all models was conducted on an NVIDIA A100 GPU with 80 GB of VRAM memory. For stage one, the models were trained using a batch size of 64, while for stage two, a batch size of 16 was used. A learning rate of $1 \times 10^{-4}$ was applied to all models, a standard choice for diffusion models to ensure stable and efficient convergence. The Adam optimizer [39] was used with $\beta_1 = 0.9$ and $\beta_2 = 0.9$, enabling the models to effectively capture complex dependencies while maintaining training stability.

## V. Results and Discussion

This section evaluates the performance of each stage of the proposed methodology, presents ablation studies to validate its benefits, and provides a discussion of the obtained results.

### A. Stage One Experiments

To evaluate the performance of the proposed method in translating RGB images to DSMs, a comprehensive experiment was conducted in stage one. The experiment aimed to compare the effectiveness of two distinct approaches—BBDMs and DDPMs—across multiple horizontal resolutions. The resolutions selected for this study were 2.0 × 2.0 m, 1.0 × 1.0 m, and 0.5 × 0.5 m, with the footprint diameter for each resolution calculated as $(2x^2)^{1/2}$, where $x$ is the horizontal resolution, to ensure complete area coverage without gaps.

For each resolution, both BBDMs and DDPMs were trained using identical input data: a coarse-resolution DSM at 6 × 3 m, derived from satellite LiDAR simulations, paired with an RGB image matching the target resolution of the desired output. This setup allowed for a direct comparison between the two methods under consistent conditions. The models were trained to generate DSMs at their respective resolutions, enabling an assessment of how well each approach handled varying levels of detail.

To provide a thorough evaluation, the experiment included a region-specific analysis, assessing model performance independently for each NEON region. This approach allowed for a detailed understanding of how well each method performed across different geographical areas.

An additional test was conducted to further explore the capabilities of the BBDM in the RGB-to-DSM translation task, evaluating its performance in super-resolving images from different years, despite being trained solely on data from a single specific year, and assessing its utility for high-resolution change detection. This test used the model trained exclusively on data from the SERC region collected in 2022, configured to generate DSMs at a horizontal resolution of 1.0 × 1.0 m. The input was the coarse-resolution DSM at 6 × 3 m from satellite LiDAR simulations, paired with RGB images at the target resolution.

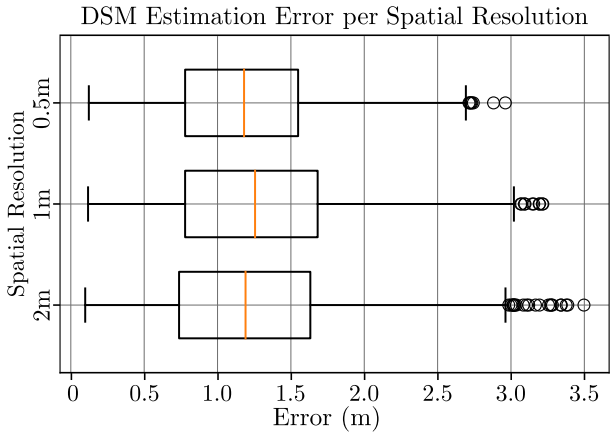



Fig. 8. MAE distribution for DSM reconstruction using the BBDM across varying horizontal resolutions. Box plots show the MAE distribution for three resolutions: (left) 2 × 2 m, (center) 1 × 1 m, and (right) 0.5 × 0.5 m. Errors are reported in meters. Coarser resolutions are easier to reconstruct due to reduced complexity but contain less spatial detail. Finer resolutions provide more detailed spatial information but are more challenging to reconstruct, with the conditioning aiding in capturing finer features.

The test pursued two main goals: first, to determine if the BBDM could generalize across time by producing accurate DSMs for years before 2022, ensuring its effectiveness across diverse temporal contexts; second, to assess whether the generated DSMs were sufficiently detailed to detect ecological changes in the SERC region, such as tree growth or loss, at a 1.0 × 1.0 m resolution.

The experiment involved applying the 2022-trained BBDM to 2017 imagery, generating corresponding high-resolution DSMs, and analyzing their accuracy and fidelity. The quality of these DSMs was assessed by comparing them to GT data and evaluating their ability to reveal fine-scale changes in vegetation structure, such as the emergence or disappearance of trees. This dual-purpose test provided insights into both the temporal robustness of the BBDM and its potential as a tool for high-resolution environmental monitoring.

### B. Stage One Results

The boxplots in Fig. 8 illustrate the distribution of DSM reconstruction errors for BBDM trained at three horizontal resolutions: 2.0 × 2.0 m, 1.0 × 1.0 m, and 0.5 × 0.5 m. At the coarsest resolution of 2.0 m, the model achieved a mean error of 1.18 m, characterized by the narrowest interquartile range (IQR) and the lowest first quartile (Q1) among the three resolutions. Only a limited number of outliers were observed, primarily resulting from artifacts in the testing regions. This coarser resolution simplifies the reconstruction process by reducing the demand for fine-scale detail generation; however, the corresponding 2.0-m RGB images provide limited structural information, potentially constraining the model's ability to recover certain features. In contrast, at the intermediate 1.0-m resolution, the mean error increased slightly to 1.28 m, accompanied by the widest IQR, as well as the highest

TABLE I

COMPARISON OF THE BBDM AND DDPM PERFORMANCE WITH THREE BENCHMARK ALGORITHMS FOR THREE DSM RESOLUTIONS

| | | MAE (m) ↓ | | | SSIM ↑ | | | PSNR ↑ | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Region | Model | 2.0 | 1.0 | 0.5 | 2.0 | 1.0 | 0.5 | 2.0 | 1.0 | 0.5 |
| UKFS | DDPM | **1.46** | 1.66 | 2.01 | **0.86** | **0.81** | 0.76 | **28.49** | **27.45** | 25.98 |
| | BBDM | 1.49 | **1.65** | **1.61** | 0.85 | **0.81** | **0.81** | 28.46 | 27.27 | **27.74** |
| SCBI | DDPM | **0.72** | 0.79 | 0.84 | **0.92** | **0.90** | 0.89 | 35.27 | 34.85 | 34.47 |
| | BBDM | 0.74 | **0.75** | **0.68** | 0.91 | **0.90** | **0.91** | **35.36** | **35.09** | **35.79** |
| SERC | DDPM | **1.31** | **1.45** | 1.40 | **0.88** | **0.83** | 0.82 | **28.85** | **28.15** | 28.25 |
| | BBDM | 1.38 | **1.45** | **1.28** | 0.87 | **0.83** | **0.84** | 28.74 | 27.94 | **28.70** |

median and Q1 values. Although this resolution produced more outliers than the 2.0-m case, these were less extreme in magnitude. The 1.0-m resolution strikes a balance between capturing finer details and relying on the structural information available in the RGB images, yet the increased complexity of resolving small-scale structures—coupled with potentially insufficient input detail—led to greater error variability. At the finest resolution of 0.5 m, the mean error returned to 1.18 m, with an IQR width intermediate to the other two cases. While the Q1 value was lower than that of the 1.0-m resolution, a higher number of more extreme outliers were observed above Q1, predominantly in regions containing small, isolated features such as power lines or water holes that deviated significantly from their surroundings. Although the 0.5-m resolution offers the potential for highly detailed DSMs, the complexity of generating fine-scale features, combined with the limited information provided by high-resolution RGB images, resulted in elevated errors in challenging areas. This analysis highlights the tradeoffs between resolution, reconstruction fidelity, and error distribution in DSM generation using BBDM models.

The performance of the BBDM for RGB-to-DSM translation is evaluated against the DDPM in Table I, using three metrics: MAE, SSIM, and PSNR, across three horizontal resolution levels: 2.0, 1.0, and 0.5 m. The evaluation was conducted on the test set of the dataset described in Section IV-C, comprising approximately 900 96 × 96 m subregions from the UKFS, the SCBI, and the SERC. The results demonstrate that the BBDM significantly outperforms the DDPM at the challenging 0.5-m resolution, achieving superior accuracy and structural fidelity critical for high-resolution DSM applications. At resolutions of 1.0 and 2.0 m, the BBDM maintains a consistent advantage, with marginal yet meaningful improvements in all metrics, underscoring its robustness and potential for advancing high-precision geospatial modeling.

Visual comparison for the three BBDM models is presented in Figs. 9 and 10 for two 96 × 96 m regions. Each figure includes the RGB image and the corresponding coarse-resolution DSM used as input to the model. In addition, the figures show the BBDM model's reconstructed DSMs and the GT DSMs for comparison. The last row in each figure contains absolute error maps, which show the pixel-wise absolute error between the reconstructions and the GT. To ensure consistency, all three error maps within a figure use the same color scale. Fig. 9 illustrates results for a vegetated area without any human-made structures, while Fig. 10 features a region with

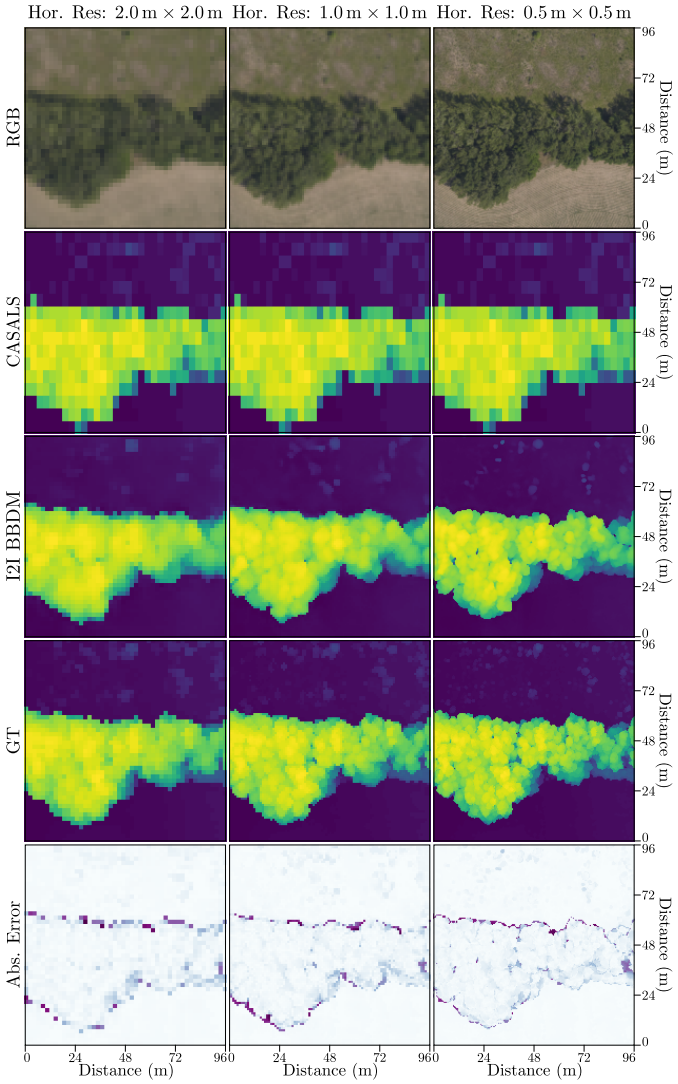



Fig. 9. Stage one results: I2I translation output for different horizontal resolutions: (left) 2.0 × 2.0 m, (center) 1.0 × 1.0 m, and (right) 0.5 × 0.5 m. (Top to bottom) RGB image used at input; conditioning DSM captured by the satellite CASALS mission at a 6 × 3 m horizontal resolution; output of the BBDM I2I translation model; GT; and absolute error map.

a mix of vegetation, large buildings, and challenging features such as power lines. These results show the capability of the BBDM model to generate high-resolution images that recover large and fine features from the region by leveraging the high-resolution RGB.

### C. Stage Two Results

Fig. 11 presents the performance evaluation of the BBDM in generating the DSM for a 576 × 576 m subset of the SERC region at a horizontal resolution of 1 × 1 m, for the years 2022 and 2017. To facilitate detailed analysis, 200 × 200 m zoomed-in view sections are provided, marked by red squares. The figure is structured into four rows: i) the first row displays RGB images at 1 × 1 m resolution for both years, serving as input References; ii) the second row shows CASALS DSMs at a coarser resolution of 6 × 3 m for the corresponding years, accompanied by a height change map that illustrates elevation differences between 2017 and 2022, with red indicating areas

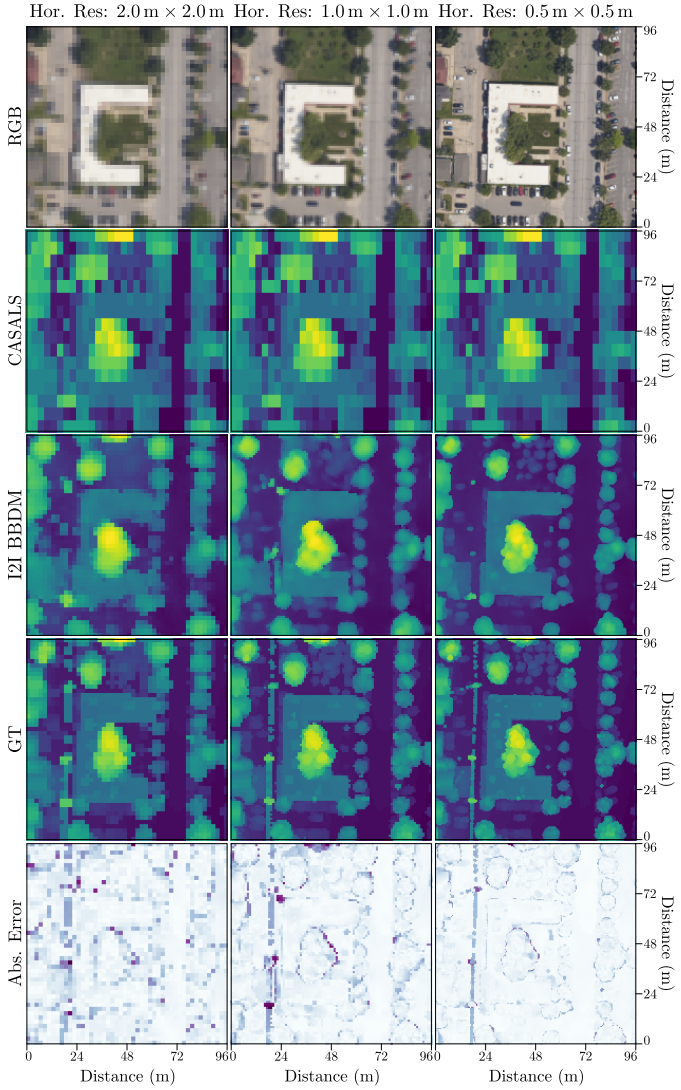



Fig. 10. Stage one results: I2I translation output for different horizontal resolutions: (left) 2.0 × 2.0 m, (center) 1.0 × 1.0 m, and (right) 0.5 × 0.5 m. (Top to bottom) RGB image used at input; conditioning DSM captured by the satellite CASALS mission at a 6 × 3 m horizontal resolution; output of the BBDM I2I translation model; GT; and absolute error map.

of height reduction and blue indicating height increase; iii) the third row presents DSMs generated by the BBDM model at 1 × 1 m resolution for both years, along with a derived height change map; and iv) the fourth row provides GT DSMs at 1 × 1 m resolution for validation, including their respective height change map. The BBDM-generated DSMs demonstrate the model's capability to effectively super-resolve coarser DSM inputs into high-resolution outputs, accurately capturing fine-scale surface features across both temporal instances. Notably, despite being trained exclusively on 2022 data, the model successfully generalizes to 2017 data, underscoring its robustness across different temporal contexts. Furthermore, the height change map derived from the BBDM outputs exhibits superior accuracy and detail compared to that obtained from the coarser CASALS DSMs, thereby enabling more precise monitoring of landscape dynamics, such as vegetation growth or loss. These results highlight the efficacy of the proposed methodology for high-resolution environmental monitoring applications.

### D. Stage Two Experiments

Stage two experiments evaluated the performance of the proposed SIM-ID method for HHDC super-resolution. SIM-ID was compared against three baseline models. The first model, super-resolution HHDC generative adversarial network (SRHHDC-GAN), is a convolutional neural network with adversarial training that takes the coarse-resolution LiDAR measurements, concatenated with the high-resolution RGB image, as input. The second model, C-DDPM, used only the coarse-resolution measurement as conditioning, without incorporating additional side information. The third model, CSI-DDPM, included the high-resolution RGB image in its conditioning by concatenating it with the coarse-resolution HHDC without incorporating the binary mask $\mathcal{M}$.

To qualitatively evaluate the performance of stage two, Fig. 12 compares a CHP reconstructed using three methods: C-DDPM, CSI-DDPM, and the proposed SIM-ID. The CHP was randomly selected from an HHDC generated from a 576 × 576 m region in SERC. SERC manages forests typical of the mid-Atlantic region. Among its tree species, the tulip poplar (Liriodendron tulipifera) is particularly notable for its height, with individuals in the eastern United States reaching 100 to 150 feet (approximately 30–45 m). These towering trees contribute to the complex canopy structures modeled in this study. Fig. 12 also includes the coarse-resolution HHDC profile, simulating data from a satellite LiDAR sensor (around 500-km AGL), and the true high-resolution HHDC profile, simulating data from an airborne LiDAR sensor (around 450-m AGL). Both coarse and high-resolution HHDC profiles were generated from point cloud simulations.

Each CHP reconstruction features a red line representing the true DSM profile and a blue line representing the reconstructed DSM profile. The difference between these lines is highlighted by the overlaid red area, referred to as the DSM error area, which indicates reconstruction errors. In this forested region, sections where the blue line deviates above or below the red line correspond to hallucinated or missing canopy structures, respectively.

The C-DDPM reconstruction exhibits the largest DSM error area, with significant inaccuracies in canopy representation, as highlighted in the green and orange boxes. These errors stem from its reliance solely on coarse-resolution HHDC data, without incorporating side information. In contrast, both CSI-DDPM and SIM-ID utilize side information for reconstruction. While CSI-DDPM reduces errors compared to C-DDPM, SIM-ID achieves superior accuracy, as evidenced by minimal DSM error areas. By effectively utilizing side information, SIM-ID is able to capture finer canopy details and minimize both hallucinated and missing structures. This highlights the robustness and accuracy of the proposed approach in reconstructing HHDCs with high fidelity, even in challenging forested environments.

### E. Ablation Studies

To evaluate the impact of incorporating side information on the fidelity of generated HHDCs, a series of ablation studies was conducted. These experiments were designed to

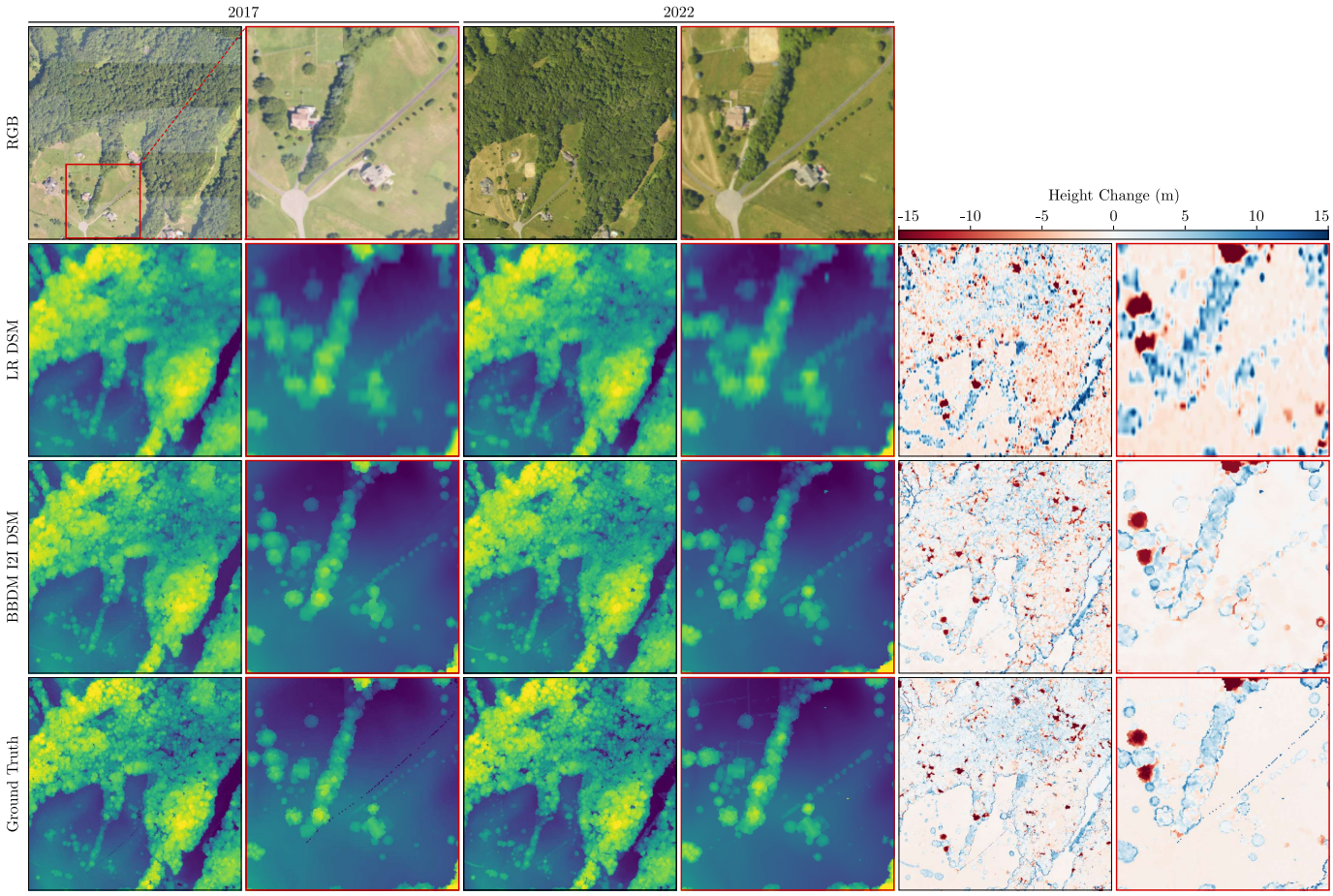



Fig. 11. Stage one results: I2I translation using BBDM for a portion of the SERC region using data from 2017 and 2022. (Left to right) Full 576 × 576 m region for 2017; zoomed-in view 200 × 200 m subregion for 2017; full region for 2022; zoomed-in view subregion for 2022; height change map for the full region; height change map for the zoomed-in view subregion. In the height change maps, blue pixels indicate areas of height increase, and red pixels indicate areas of height decrease between 2017 and 2022. (Top to bottom) RGB images at 1-m horizontal resolution; CASALS DSMs and their height change maps; BBDM 1-m super-resolved DSMs and their height change maps; and GT DSMs and their height change maps.

validate the hypothesis that the inclusion of side information enhances the quality of the generated HHDCs, while also demonstrating that the specific methodology proposed in this article further improves fidelity and reduces hallucinations compared to alternative approaches. All models were trained and assessed under identical experimental conditions to ensure a fair and consistent comparison.

A baseline model, called C-DDPM, was trained using exclusively the low-resolution HHDC as input. This configuration isolates the performance achievable without side information. As shown in Fig. 12, the C-DDPM exhibits higher error rates compared to models that leverage additional data sources, serving as a reference point to quantify the benefits of side information.

To further investigate the role of side information and validate the proposed methodology, two additional models were trained, both receiving the same set of inputs: the RGB image, the high-resolution DSM, and the low-resolution HHDC. These inputs were processed differently by each model to assess the effectiveness of the proposed approach: The CSI-DDPM model concatenates the RGB image, the high-resolution DSM, and the low-resolution HHDC into a single tensor. The resulting tensor is then utilized as conditioning for the DDPM. This straightforward approach

TABLE II
COMPARISON OF THE PROPOSED APPROACHES WITH THREE BENCHMARK ALGORITHMS FOR 1-M SUPER-RESOLUTION ON SERC

| 1-meter Resolution | $\text{PSNR}_{\text{chp}} \uparrow$ | $\text{MSE}_{\text{chp}} \downarrow$ | $\text{SSIM}_{chp} \uparrow$ |
|---|---|---|---|
| SRHHDC-GAN | 16.5471 | 0.0222 | 0.5573 |
| C-DDPM | 16.9255 | 0.0204 | 0.5857 |
| CSI-DDPM | 17.0836 | 0.0196 | 0.5942 |
| SIM-ID | **17.5741** | **0.0175** | **0.6244** |

serves as a comparative baseline for integrating side information. On the other hand, the SIM-ID model follows the methodology proposed in this article, processing the same three inputs—RGB image, high-resolution DSM, and low-resolution HHDC—through a specialized mechanism, as detailed in Section Stage two: Constrained Diffusion Model. The SIM-ID approach is designed to optimally leverage side information, enhancing the generation process.

The experimental results, presented in Section V-C, confirm that the inclusion of side information significantly improves the fidelity of the generated high-resolution HHDCs. Specifically, the SIM-ID model outperforms traditional super-resolution models like SRHHDC-GAN, and newer diffusion-based models like C-DDPM, and CSI-DDPM in terms of PSNR, MSE, and SSIM on the generated CHPs

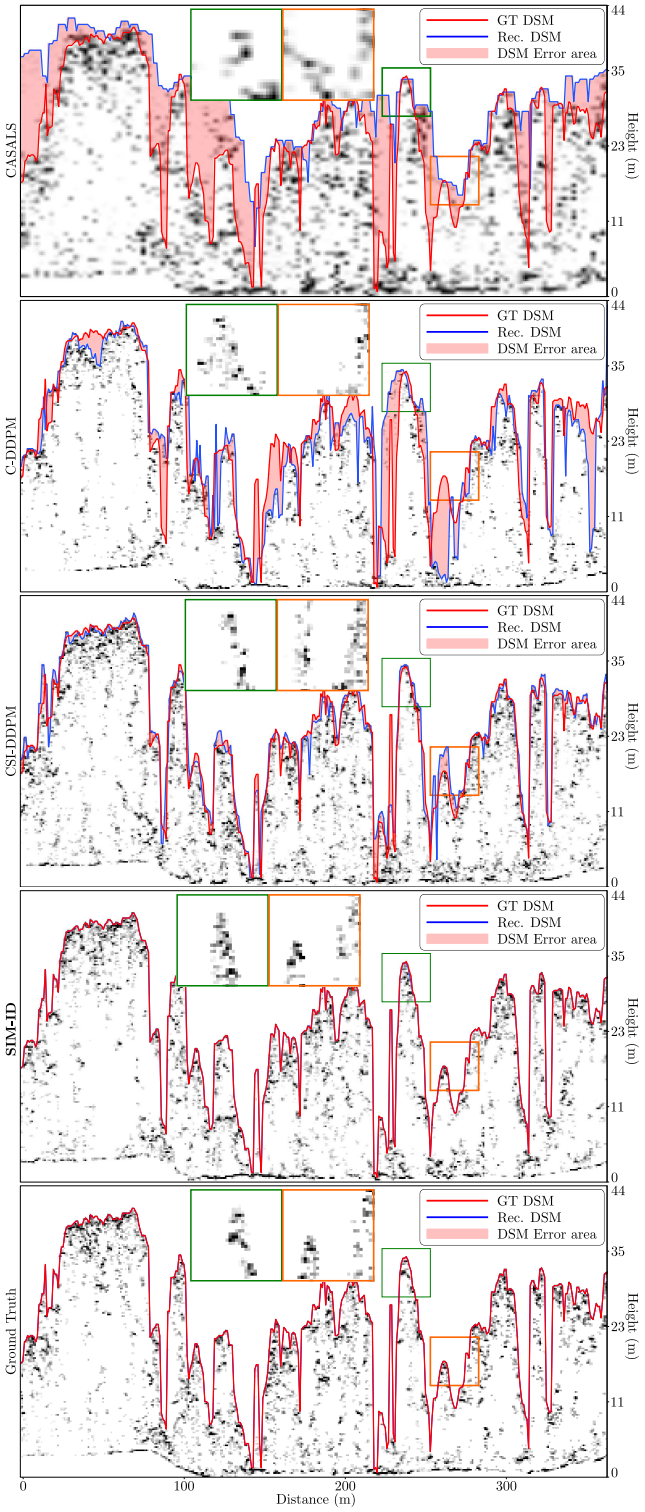



Fig. 12. Stage two results: comparison of CHP reconstructions using C-DDPM, CSI-DDPM, and the proposed SIM-ID. The CHP was randomly selected from an HHDC generated for a region in SERC. (Top to bottom) Coarse-resolution CHP (simulating satellite LiDAR) C-DDPM, CSI-DDPM, SIM-ID reconstructions, and the true high-resolution CHP (simulating airborne LiDAR data). Each reconstruction is shown with the true DSM profile (red line) and the reconstructed DSM profile (blue line), with reconstruction errors highlighted as the DSM error area (red-shaded regions). C-DDPM exhibits significant errors due to reliance on coarse-resolution data, while CSI-DDPM reduces these errors with side information, and SIM-ID achieves the most accurate reconstruction with minimal DSM error areas.

as seen in Table II. Compared to the baseline C-DDPM, both diffusion-based models using side information reduce error rates; however, the SIM-ID model demonstrates a marked improvement over the CSI-DDPM, exhibiting fewer hallucinations and higher fidelity in the generated HHDCs. These findings validate that the proposed methodology for processing side information is more effective than a naive concatenation approach.

## VI. Conclusion

This article introduces an advanced methodology for super-resolving DSMs and HHDCs by leveraging high-resolution monocular RGB imagery and coarse-resolution satellite LiDAR measurements. Both cases are based on recent generative AI techniques, namely, DDPMs and BBDMs. This methodology is designed in a two-stage fashion. The first stage uses specialized techniques to translate the high-resolution RGB information to the DSM domain, giving access to the high-resolution canopy structure. The second stage defines a novel way of conditioning the HHDC reconstruction on the high-resolution canopy structure. Experiments showed how the proposed technique outperforms other methods of conditioning.

Stage one of the proposed methodology showed great adaptability to the availability of the data, maintaining the reconstruction quality as the RGB horizontal resolution changes. The second stage of the proposed methodology proved to increase the quality of the HHDC reconstructions and reduce the model's hallucinations, which is a critical factor when using generative AI techniques. This artifact and hallucination reduction increases the usability of the reconstruction on critical applications like hydrology and bare Earth studies. It is worth noting that the achievable spatial resolution depends on the resolution of the monocular RGB images in the observation system, the density of photons in the training point cloud, and the level of noise in the LiDAR sensor.

For practical deployment, it is essential to have co-registered satellite LiDAR measurements and high-resolution RGB images. For global-scale measurements, this requirement can be addressed by combining data from high-resolution sources such as WorldView-3, which provides panchromatic images with a horizontal resolution of up to 0.3 m [17]. However, to ensure accuracy, the satellite and RGB data must be collected within a similar timeframe, and the terrain must be free of clouds or other types of atmospheric noise. Another limitation relates to the computational and time costs involved in the reconstruction process. In this study, reconstructing a 576 $\times$ 576 m region at a 1 $\times$ 1 m horizontal resolution requires approximately 58 min on an NVIDIA A100 GPU with 80 GB of memory. While this performance is suitable for research purposes, it may limit the feasibility of real-time data streaming. Scaling the methodology for larger areas or achieving faster processing times would require further optimization of the computational pipeline or the use of more powerful hardware.

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